Who are the most active doctors and are they helping our product?

Your starting universe is only the doctors who appear in the Alerts dataset i.e., doctors for whom at least one alert has been generated. Do not include any doctor who has no alert record. From this universe, produce two separate Top 10 lists using two different ranking metrics:

dummy DataFrame as df

```
-- 1.1

-- Ranking 1 - Top 10 by Alert Volume: Which 10 doctors have the highest total number of alerts generated?

with alert_counts as (
    select
        hcp_id,
        count(alert_id) as total_alerts,
        row_number() over (order by count(alert_id) desc) as rank
    from
        Alerts.csv
    group by
        hcp_id
)
select
    rank,
    hcp_id,
    total_alerts
from
    alert_counts
where
    rank <= 10;
```

index	rank	HCP_ID	total_alerts
0	1	HCP041	
1	2	HCP011	
2	3	HCP019	
3	4	HCP020	
4	5	HCP032	
5	6	HCP021	
6	7	HCP014	
7	8	HCP036	
8	9	HCP022	
9	10	HCP007	

Rows: 10 Expand

dummy DataFrame as df1

```
-- 1.2

-- Ranking 2 - Top 10 by Prescription Volume: Which 10 doctors from the alerts universe have written the highest prescription
volume in the Sales data?

with filtered_sales as (
    select
        s.hcp_id,
        sum(s.prescription_volume) as total_prescription_volume,
        row_number() over (order by sum(s.prescription_volume) desc) as rank
    from
        Sales.csv s
    inner join (
        select distinct hcp_id
        from Alerts.csv
    ) a on s.hcp_id = a.hcp_id
    group by
        s.hcp_id
)
select
    rank,
    hcp_id,
    total_prescription_volume
from
    filtered_sales
where
    rank <= 10;
```

index	rank	HCP_ID	total_prescription_volume
0	1	HCP185	
1	2	HCP062	
2	3	HCP019	
3	4	HCP163	
4	5	HCP211	
5	6	HCP126	
6	7	HCP154	
7	8	HCP215	
8	9	HCP088	
9	10	HCP004	

Rows: 10 Expand

Which doctors see the need but don't act on it?

dummy DataFrame as df2

```
-- 2

-- Identify all doctors from the alerts universe who received positive lab test alerts, indicating that the patient likely required treatment. Evaluate whether these doctors showed any prescription activity for any drug within 30 days following their first positive alert. Among doctors with no prescribing activity during this post-alert window, rank the Top 10 HCPs by total positive alert volume.


with positivealerts as (
    select
        hcp_id,
        count(alert_id) as total_positive_alerts,
        min(alert_date) as first_positive_alert_date
    from Alerts.csv
    where lower(lab_result) = 'positive'
    group by hcp_id
),
prescriptionswithin30days as (
    select distinct
        p.hcp_id
    from positivealerts p
    join Sales.csv s
        on p.hcp_id = s.hcp_id
    where s.prescription_date >= p.first_positive_alert_date
      and s.prescription_date <= p.first_positive_alert_date + interval '30 days'
)
select
    row_number() over (order by pa.total_positive_alerts desc) as rank,
    pa.hcp_id,
    pa.total_positive_alerts as "positive alerts"
from positivealerts pa
left join prescriptionswithin30days rx
    on pa.hcp_id = rx.hcp_id
where rx.hcp_id is null
order by pa.total_positive_alerts desc
limit 10;
```

...	...	...	positive al...
0	1	HCP041	127
1	2	HCP011	123
2	3	HCP021	116
3	4	HCP042	101
4	5	HCP023	99
5	6	HCP007	96
6	7	HCP026	78
7	8	HCP009	67
8	9	HCP001	63
9	10	HCP002	59

Rows: 10 Expand

Among our top doctors, how much business is going to competitors?

dummy DataFrame as d

```
-- 3.1

/*
Use the Top 10 doctors by total alert volume from Question 1.
For these 10 doctors, calculate the share of total prescription volume contributed by:
Our Drug (DRG001)
Competitor Drugs (all other drugs)
*/

with top10alerthcps as (
    select
        hcp_id
    from Alerts.csv
    group by hcp_id
    order by count(alert_id) desc
    limit 10
),
hcp_mix as (
    select
        t.hcp_id,
        coalesce(sum(s.prescription_volume), 0) as total_prescription_volume,
        coalesce(sum(case
            when lower(s.drug_id) = 'drg001'
            then s.prescription_volume
            else 0
        end), 0) as our_drug_volume,
        coalesce(sum(case
            when lower(s.drug_id) <> 'drg001'
            then s.prescription_volume
            else 0
        end), 0) as competitor_volume
    from top10alerthcps t
    left join Sales.csv s
        on t.hcp_id = s.hcp_id
    group by t.hcp_id
)
select
    hcp_id,
    total_prescription_volume,
    case
        when total_prescription_volume = 0
        then '0.0%'
        else round(our_drug_volume * 100.0 / total_prescription_volume, 1)::text || '%'
    end as "% our drug share",
    case
        when total_prescription_volume = 0
        then '0.0%'
        else round(competitor_volume * 100.0 / total_prescription_volume, 1)::text || '%'
    end as "% competitor share"
from hcp_mix
order by
    case
        when total_prescription_volume = 0 then 0
        else competitor_volume * 100.0 / total_prescription_volume
    end desc;
```

...	...	total_prescription_volume	% our drug ...	% competitor s...
0	HCP020	570	25.4%	74.6%
1	HCP019	2277	59.1%	40.9%
2	HCP032	2075	59.1%	40.9%
3	HCP022	1363	62.0%	38.0%
4	HCP036	1895	64.7%	35.3%
5	HCP014	944	71.1%	28.9%
6	HCP021	0	0.0%	0.0%
7	HCP007	0	0.0%	0.0%

dummy DataFrame as

```sql
-- 3.2

-- Identify which competitor drug accounts for the highest prescription volume and is capturing the largest share among your highest-alerting doctors.

with top10alerthcps as (
    select
        hcp_id
    from Alerts.csv
    group by hcp_id
    order by count(alert_id) desc, hcp_id
    limit 10
)
select
    s.drug_name as "competitor drug name",
    s.drug_id,
    sum(s.prescription_volume) as total_prescription_volume
from Sales.csv s
join top10alerthcps t
    on s.hcp_id = t.hcp_id
where lower(s.drug_id) <> 'drg001'
group by
    s.drug_name,
    s.drug_id
order by
    sum(s.prescription_volume) desc,
    s.drug_id
limit 1;
```

	competitor drug n...		total_prescription_volume
0	Rivalex	DRG002	1470

Rows: 1 Expand

Which hospitals are sitting on the biggest untapped opportunity?

dummy DataFrame as

```
-- 4

/*
Using the Affiliation dataset, identify the Top 10 accounts with the highest alert volume based on activity generated by their
affiliated doctors.
For these Top 10 accounts, calculate:
Total Alert Count
Our Drug Prescription Count (DRG001)
Conversion Ratio - defined as the proportion of prescription count relative to alert count
Rank the accounts by lowest conversion ratio to identify high-alert accounts with weak prescription conversion.
*/

with accountalerts as (
    select
        af.account_id,
        count(al.alert_id) as total_alerts
    from Alerts.csv al
    inner join Affiliation.csv af
        on al.hcp_id = af.hcp_id
    group by af.account_id
),
top10accounts as (
    select
        account_id,
        total_alerts
    from accountalerts
    order by total_alerts desc, account_id
    limit 10
),
accountprescriptions as (
    select
        af.account_id,
        count(s.prescription_id) as our_drug_rx_count
    from Sales.csv s
    inner join Affiliation.csv af
        on s.hcp_id = af.hcp_id
    where lower(s.drug_id) = 'drg001'
    group by af.account_id
)
select
    row_number() over (
        order by coalesce(rx.our_drug_rx_count, 0) * 1.0 / aa.total_alerts asc
    ) as rank,
    aa.account_id,
    aa.total_alerts as "total alerts",
    coalesce(rx.our_drug_rx_count, 0) as "our drug prescription count",
    round(
        coalesce(rx.our_drug_rx_count, 0) * 100.0 / aa.total_alerts,
        1
    )::text || '%' as "conversion ratio"
from top10accounts aa
left join accountprescriptions rx
    on aa.account_id = rx.account_id
order by rank;
```

		A...	total ...	our drug prescription count	conversion r...
0	1	ACC001	3296	385	11.7%
1	2	ACC003	1195	159	13.3%
2	3	ACC002	2350	373	15.9%
3	4	ACC004	530	164	30.9%
4	5	ACC005	585	275	47.0%
5	6	ACC006	493	298	60.4%
6	7	ACC012	85	57	67.1%
7	8	ACC009	100	77	77.0%
8	9	ACC007	114	209	183.3%

dummy DataFrame as

```
-- 5

/*
Using the Affiliation dataset, determine the total number of doctors associated with each account. Using prescription activity
for Our Drug, calculate:
The number of active prescribers within each account
Total prescription volume at the account level
Finally, rank accounts by the lowest prescriptions-per-active-doctor ratio and identify the Top 10 accounts.
*/

with accountbase as (
    select
        account_id,
        count(distinct hcp_id) as total_affiliated_doctors
    from Affiliation.csv
    group by account_id
),
activeprescribers as (
    select
        a.account_id,
        count(distinct s.hcp_id) as active_doctors,
        sum(s.prescription_volume) as total_rx_volume
    from Sales.csv s
    inner join Affiliation.csv a
        on s.hcp_id = a.hcp_id
    where lower(s.drug_id) = 'drg001'
    group by a.account_id
)
select
    row_number() over (
        order by (ap.total_rx_volume * 1.0 / ap.active_doctors) asc
    ) as rank,
    ab.account_id,
    ab.total_affiliated_doctors as "# affiliated doctors",
    ap.active_doctors as "# active doctors",
    ap.total_rx_volume as "our drug prescription volume",
    round(ap.total_rx_volume * 1.0 / ap.active_doctors, 1)
        as "prescription volume per active doctor"
from accountbase ab
join activeprescribers ap
    on ab.account_id = ap.account_id
where ap.active_doctors > 0
order by "prescription volume per active doctor" asc
limit 10;
```

...	...	A...	# affiliated doctors	# active do...	our drug prescription volume	prescription volume p
0	1	ACC017	3	1	59	
1	2	ACC015	8	7	1347	
2	3	ACC012	10	7	1804	
3	4	ACC009	11	10	2699	
4	5	ACC003	24	15	4947	
5	6	ACC018	4	3	1095	
6	7	ACC004	17	13	5402	
7	8	ACC008	7	6	3049	
8	9	ACC014	6	6	3126	
9	10	ACC023	1	1	573	

Rows: 10 Expand

Did alerts actually change doctor behavior - before vs. after?

dummy DataFrame as

```
-- 6.1

-- Part A - Basic lift: Define a 90-day before and 90-day after window around each doctor's first positive alert. Calculate prescription volume for our drug in each window. Rank Top 10 HCPs by lift % with highest lift being Rank 1?

with firstalerts as (
    select
        hcp_id,
        min(alert_date) as first_alert_date
    from Alerts.csv
    where lower(lab_result) = 'positive'
    group by hcp_id
),
saleswindows as (
    select
        fa.hcp_id,
        coalesce(sum(
            case
                when s.prescription_date >= fa.first_alert_date - interval '90 days'
                 and s.prescription_date < fa.first_alert_date
                then s.prescription_volume
                else 0
            end
        ), 0) as pre_volume,
        coalesce(sum(
            case
                when s.prescription_date >= fa.first_alert_date
                 and s.prescription_date <= fa.first_alert_date + interval '90 days'
                then s.prescription_volume
                else 0
            end
        ), 0) as post_volume
    from firstalerts fa
    left join Sales.csv s
        on fa.hcp_id = s.hcp_id
       and lower(s.drug_id) = 'drg001'
    group by fa.hcp_id
),
liftcalculation as (
    select
        hcp_id,
        pre_volume,
        post_volume,
        case
            when pre_volume > 0
            then ((post_volume * 1.0 - pre_volume) / pre_volume) * 100
            else null
        end as lift_pct
    from saleswindows
)
select
    row_number() over (order by lift_pct desc) as rank,
    hcp_id,
    pre_volume as "total prescription volume before alert",
    post_volume as "total prescription volume after alert",
    '+' || round(lift_pct, 1)::text || '%' as "lift %"
from liftcalculation
where lift_pct is not null
order by lift_pct desc
limit 10;
```

...	...	...	total prescription volume before alert	total prescription volume after alert	...
0	1	HCP040	8	143	+1687.5%
1	2	HCP050	10	107	+970.0%
2	3	HCP006	32	249	+678.1%
3	4	HCP108	31	238	+667.7%

Rows: 10 Expand

dummy DataFrame as

```sql
-- 6.2

-- Part B - Segment the lift: Split all doctors into 3 behavioral bucket as mentioned in the formula table. Does the lift look different across the three groups? Which group shows strongest commercial impact?

with firstalerts as (
    select
        hcp_id,
        min(alert_date) as first_alert_date
    from Alerts.csv
    where lower(lab_result) = 'positive'
    group by hcp_id
),
saleswindows as (
    select
        fa.hcp_id,
        coalesce(sum(
            case
                when s.prescription_date >= fa.first_alert_date - interval '90 days'
                 and s.prescription_date < fa.first_alert_date
                then s.prescription_volume
                else 0
            end
        ), 0) as pre_volume,
        coalesce(sum(
            case
                when s.prescription_date >= fa.first_alert_date
                 and s.prescription_date <= fa.first_alert_date + interval '90 days'
                then s.prescription_volume
                else 0
            end
        ), 0) as post_volume
    from firstalerts fa
    left join Sales.csv s
        on fa.hcp_id = s.hcp_id
       and lower(s.drug_id) = 'drg001'
    group by fa.hcp_id
),
hcp_segments as (
    select
        hcp_id,
        pre_volume,
        post_volume,
        (post_volume - pre_volume) as absolute_growth,
        case
            when pre_volume > 0
            then ((post_volume * 1.0 - pre_volume) / pre_volume) * 100
            else null
        end as lift_pct,
        case
            when pre_volume = 0
                 and post_volume > 0
            then 'new starters'
            when pre_volume > 0
                 and ((post_volume * 1.0 - pre_volume) / pre_volume) * 100 >= 20
            then 'growers'
            else 'non-responders'
        end as segment
    from saleswindows
)
select
    coalesce(segment, 'total') as segment,
    count(hcp_id) as hcps,
    case
        when avg(lift_pct) is null
        then 'n/a'
        else '+' || round(avg(lift_pct), 1)::text || '%'
    end as "avg lift",
```

```
order by
case
    when segment is null then 1
    else 0
end,
segment;
```

	segment			total absolute growth (commercial i...
0	growers	25	+322.8%	3319
1	new starters	64	n/a	9516
2	non-responders	125	+-68.2%	-4623
3	total	214	+69.5%	8212

Rows: 4 Expand

Commercial HCP Prioritization

dummy DataFrame as

```
-- 7.1

/*
A pharma company assign sales representatives to promote the drug to HCP. Your company’s commercial team is planning the next quarter’s outreach strategy for DRG001. Following are the field force assumptions:
Each sales rep can complete ~8 effective HCP calls per day
Each rep works ~20 field days per month
The organization has 40 sales reps nationally
Leadership wants to use this limited field force capacity efficiently by prioritizing the HCP segments identified in Question 6 (New Starters, Growers, and Non-Responders).
Using the segment-level insights from Question 6, recommend how the company should allocate total rep call capacity across these segments.
In your response:
Estimate how much outreach focus each segment should receive
Explain the reasoning behind your allocation approach
Discuss the tradeoff between growth opportunity, conversion likelihood, and sales effort required across segments
*/

with params as (
    select
        40 as total_reps,
        20 as field_days_per_month,
        3 as months_in_quarter,
        8 as calls_per_day,
        0.45 as new_starter_weight,
        0.15 as grower_weight,
        0.40 as non_responder_weight
),
capacity as (
    select *,
           total_reps * field_days_per_month * months_in_quarter * calls_per_day
           as total_capacity
    from params
),
segmentdata as (
    select 'new starters' as segment, 64 as hcp_count
    union all
    select 'growers', 25
    union all
    select 'non-responders', 125
)
select
    segment,
    hcp_count as "hcps",
    case
        when segment = 'new starters'
            then round(new_starter_weight * 100, 0)::text || '%'
        when segment = 'growers'
            then round(grower_weight * 100, 0)::text || '%'
        when segment = 'non-responders'
            then round(non_responder_weight * 100, 0)::text || '%'
    end as "suggested % of rep calls",
    case
        when segment = 'new starters'
            then cast(total_capacity * new_starter_weight as integer)
        when segment = 'growers'
            then cast(total_capacity * grower_weight as integer)
        when segment = 'non-responders'
            then cast(total_capacity * non_responder_weight as integer)
    end as "estimated quarterly calls",
    case
        when segment = 'new starters'
            then cast(round(total_reps * new_starter_weight, 0) as integer)
        when segment = 'growers'
            then cast(round(total_reps * grower_weight, 0) as integer)
        when segment = 'non-responders'
            then cast(round(total_reps * non_responder_weight, 0) as integer)
    end as "estimated rep allocation"
```

```
    max(total_capacity),
    max(total_reps)
from segmentdata
cross join capacity;
```

	segment		suggested % of rep calls	estimated quarterly calls	estimated rep allocation
0	new starters	64	45%	8640	18
1	non-responders	125	40%	7680	16
2	growers	25	15%	2880	6
3	total	214	100%	19200	40

Rows: 4 Expand

dummy DataFrame as

```
-- 7.2

-- Total rep call capacity per quarter

with params as (
    select
        40 as total_reps,
        20 as field_days_per_month,
        3 as months_in_quarter,
        8 as calls_per_day
)
select
    total_reps * field_days_per_month * months_in_quarter * calls_per_day
    as "total rep call capacity per quarter"
from params;
```

	total rep call capacity per quarter
0	19200

Rows: 1 Expand